

Shamsa Abid, Ph.D.

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 <https://shamsaabid.github.io/>

Research Interests

- AI and Deep Learning systems Explainability, Trustworthiness, Generative AI for Code Reuse - Safety and Performance, Code Models Trustworthiness Evaluation, Semantic Code Clones Benchmarking, Explainable AI (XAI), Code Recommendation and Reuse, Mining Software Repositories, Empirical Studies, Design Patterns, Algorithms.

PostDoc Employment

- Oct 2025 – Present  **Research Fellow** Lahore University of Management Sciences (LUMS)
- Oct 2024 – Aug 2025  **Assistant Professor** National University of Computer and Emerging Sciences (NUCES)
- July 2024 – Dec 2024  **Post Doctoral Research Fellow** Lahore University of Management Sciences (LUMS)
- May 2022 – May 2024  **Research Scientist** Singapore Management University
- Dec 2021 – Mar 2022  **Intern** JetBrains

Education

- 2015 – 2021  **Ph.D. Computer Science, LUMS**
Thesis title: *Feature-driven API Usage based Code Recommendation for Opportunistic Reuse*
- 2011 – 2013  **MS Computer Science, LUMS**
Project title: *Low Literacy Keyboard for Mobile Devices: Enabling Fast and Efficient Text Input.*
- 2000 – 2004  **BS Hons. Computer Science, Lahore College for Women University**
Project title: *Library Management System.*

Teaching Experience

- Spring 2025  **Assistant Professor** *Software Reengineering*, NUCES
-  **Assistant Professor** *Design and Analysis of Algorithms*, NUCES
- Fall 2024  **Assistant Professor** *Software Requirements Engineering*, NUCES
-  **Assistant Professor** *Data Structures*, NUCES
- Spring 2015  **Visiting Lecturer** *Software Quality Assurance*, Kinnaird College for Women University
- Fall 2012  **Visiting Lecturer** *Creating Digital Content*, Beaconhouse National University
- Spring 2021  **Teaching Assistant** *Software Development Tools and Processes*, LUMS
- Spring 2020  **Teaching Assistant** *Software Development Tools and Processes*, LUMS
- Spring 2015  **Teaching Assistant** *Software Engineering*, LUMS
- Fall 2014  **Teaching Assistant** *Design Patterns and Refactoring*, LUMS
- Fall 2013  **Teaching Assistant** *Design Patterns and Refactoring*, LUMS
- Spring 2013  **Teaching Assistant** *Software Engineering*, LUMS
- Fall 2012  **Teaching Assistant** *Introduction to Computing*, LUMS

Industry Experience

- Aug 13 – Oct 14  **Senior Software Engineer** Techlogix Pvt. Ltd.
Apr 07 – Sep 07  **Software Engineer** Xavor Pvt. Ltd.

Research Experience

- Jan 13 – Jun 13  **Research Associate** *Software Testing* LUMS
Jun 11 – Aug 11  **Research Associate** *Technology for Low Literacy Segment*, LUMS

Research Publications

Journal Articles

- 1 S. Abid, X. Cai, and L. Jiang, “Measuring model alignment for code clone detection using causal interpretation,” *Empirical Software Engineering*, vol. 30, no. 2, p. 46, 2025.
- 2 S. Abid, S. Shamail, H. A. Basit, and S. Nadi, “FACER: An API usage-based code-example recommender for opportunistic reuse,” *Empirical Software Engineering*, vol. 26, no. 6, p. 110, 2021.

Conferences and Workshops

- 1 M. T. Jamil, S. Abid, and S. Shamail, “Can llms generate higher quality code than humans? an empirical study,” in *2025 IEEE/ACM 22nd International Conference on Mining Software Repositories (MSR)*, 2025, pp. 478–489.  DOI: 10.1109/MSR66628.2025.00081.
- 2 S. Abid, X. Cai, and L. Jiang, “Interpreting codebert for semantic code clone detection,” in *2023 30th Asia-Pacific Software Engineering Conference (APSEC)*, IEEE, 2023, pp. 229–238.
- 3 S. Abid, H. Abdul Basit, and S. Shamail, “Context-aware code recommendation in intellij idea,” in *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, 2022, pp. 1647–1651.
- 4 S. Arshad, S. Abid, and S. Shamail, “Codebert for code clone detection: A replication study,” in *2022 IEEE 16th International Workshop on Software Clones (IWSC)*, IEEE, 2022, pp. 39–45.
- 5 S. M. Rabbani, N. A. Gulzar, S. Arshad, S. Abid, and S. Shamail, “A comparative analysis of clone detection techniques on semanticclonebench,” in *2022 IEEE 16th International Workshop on Software Clones (IWSC)*, IEEE, 2022, pp. 16–22.
- 6 M. Kamal, A. Abaid, S. Abid, and S. Shamail, “Facer-as: An api usage-based code recommendation tool for android studio,” in *2021 IEEE International Conference on Software Maintenance and Evolution (ICSME)*, IEEE, 2021, pp. 609–613.
- 7 S. Abid, “Recommending related functions from api usage-based function clone structures,” in *Proceedings of the 2019 27th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, ser. ESEC/FSE 2019, Tallinn, Estonia: Association for Computing Machinery, 2019, pp. 1193–1195, ISBN: 9781450355728.  DOI: 10.1145/3338906.3342486.
- 8 S. Abid, S. Javed, M. Naseem, S. Shahid, H. A. Basit, and Y. Higo, “Codeease: Harnessing method clone structures for reuse,” in *2017 IEEE 11th International Workshop on Software Clones (IWSC)*, IEEE, 2017, pp. 1–7.
- 9 S. Abid and H. A. Basit, “Towards a structural clone based recommender system,” in *2016 IEEE 23rd International Conference on Software Analysis, Evolution, and Reengineering (SANER)*, IEEE, vol. 3, 2016, pp. 51–52.

- 10 S. Abid, Z. Qamar, N. Khan, M. Shayan, and H. A. Basit, "Retrieving design pattern usage examples using domain matching," in *2016 23rd Asia-Pacific Software Engineering Conference (APSEC)*, IEEE, 2016, pp. 33–40.
- 11 S. Abid, H. Abdul Basit, and N. Arshad, "Reflections on teaching refactoring: A tale of two projects," in *Proceedings of the 2015 ACM Conference on Innovation and Technology in Computer Science Education*, ser. ITiCSE '15, Vilnius, Lithuania: Association for Computing Machinery, 2015, pp. 225–230, ISBN: 9781450334402.  DOI: 10.1145/2729094.2742617.

Seminars

- Nov 2023  *Why P not Q? Interpreting Code Models Decision Making*, LUMS
- Oct 2023  *Unveiling the Code Model Enigma: Interpreting the Gap Between AI and Human Intuition in Code Clone Detection*, Singapore Management University
- Sep 2021  *Comparing Clustering against Deep-Learning for Semantic Code Clone Detection*, LUMS
-  *An API Usage-based Code Recommendation Tool for Android Studio*, LUMS

Service

Program Committee Membership

- 2025  **Research Track** 40th IEEE/ACM International Conference on Automated Software Engineering, ASE 2025
-  **Demonstrations Track** ACM International Conference on the Foundations of Software Engineering (FSE) 2025
-  **Research Track** International Conference on Software Maintenance and Engineering (IC-SME) 2025
- 2024  **Research Track** International Conference on Software Maintenance and Engineering (IC-SME)
-  **Technical Track** Mining Software Repositories (MSR)
-  **Research Track** AI Foundation Models and Software Engineering (Forge)
- 2023  **Demonstrations Track** International Conference on Software Engineering (ICSE)
-  **Industry Track** Mining Software Repositories (MSR)
-  **Research Track** International Workshop on Interpretability and Robustness in Neural Software Engineering (InteNSE)
- 2021  **Shadow PC** Mining Software Repositories (MSR)

Journal Reviews

- 2024  *ACM Transactions on Software Engineering and Methodology (TOSEM)*
-  *Empirical Software Engineering (EMSE)*
- 2020 – 2021  *Journal of Behavior and Information Technology*

Awards and Certificates

Awards

- 2025  **Distinguished Reviewer Award**, Demonstrations Track, ACM, FSE 2025
- 2016  **Runner Up Poster**, SBASSE PhD Colloquium, LUMS

Awards and Certificates (continued)

- 2013  **Deans Honor List Award**, LUMS
  **Outstanding Teaching Assistant Award**, Software Engineering, LUMS

Certification

- 2016  **Machine Learning Foundations: A Case Study Approach**. Coursera, University of Washington.
- 2012  **Intelligent Computing**, Course Audit Certificate, LUMS.
  **iOS Development**, Workshop Participation Certificate, LUMS.
- 2005  **FOSSAC 2005, 1st National Free and Open Source Software Awareness Campaign Training**, FOSSFP: Free and Open Source Foundation Pakistan (FOSS Academy).
- 2004  **Internet and Multimedia Professional**, BrainNET.

References

Dr. Qamar Zaman, *Assistant Professor*, HoD Software Engineering, NUCES CFD Campus, qamar.zaman@nu.edu.pk

Prof. Dr. Shafay Shamil, *Professor*, LUMS, sshamil@lums.edu.pk

Prof. Dr. Lingxiao Jiang, *Professor*, Singapore Management University, lxjiang@smu.edu.sg

Dr. Sarah Nadi, *Associate Professor*, New York University, Abu Dhabi, sarah.nadi@nyu.edu